

NYCU-ECE DCS-2026

HW03

Design: Elevator Controller

Data Preparation

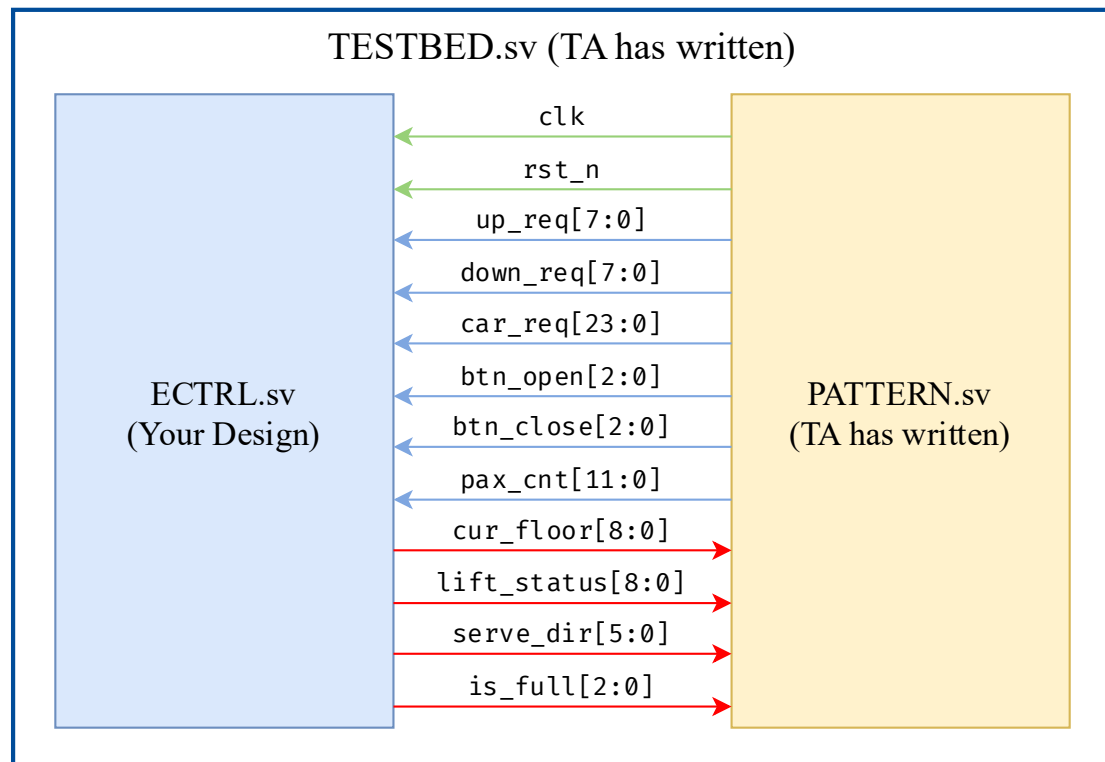
1. Extract the files needed from TA's directory:

```
$ tar -xvf ~es26dcsTA01/HW03.tar
```

2. The extracted files contain:

```
HW03/  
├── 00_TESTBED/  
├── 01_RTL/  
├── 02_SYN/  
├── 03_GATE/  
└── 09_UPLOAD/
```

Block Diagram



Design Description

In this assignment, you need to implement a multi-elevator scheduling controller (ECTRL) for a building equipped with **3 elevators (E0, E1, E2)** serving **8 floors (B1~7F)**.

1. Interface and Signals

Elevator Indexing: To manage multiple elevators, signals such as `car_req`, `btn_open`, `btn_close`, `pax_cnt`, `cur_floor`, `lift_status`, `serve_dir`, and `is_full`, are concatenated into 1D vectors. The data for the i -th elevator occupies the slice $[(i + 1)W - 1:iW]$, where W is the bit width per elevator.

For example, `car_req` is a 24-bit input; `car_req[7:0]` belongs to E0, `car_req[15:8]` belongs to E1, and so forth.

2. State Machine Definition

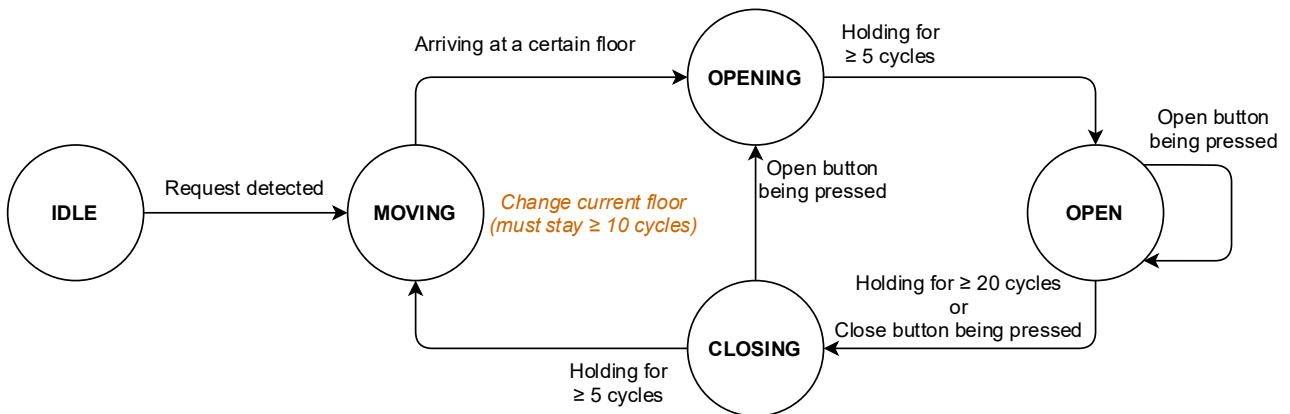
Each elevator must maintain its own operational state (`lift_status`). The physical actions follow these rules:

- **000 (IDLE)**: The initial status. The elevator is stationary with doors fully closed.
- **001 (MOVING)**: The elevator is moving. It takes **at least 10 cycles** to travel between adjacent floors.
- **010 (OPENING)**: The doors are opening. This process takes **at least 5 cycles**.
- **011 (OPEN)**: The doors are fully open, allowing passengers to board or alight. The elevator must remain in this state for **at least 20 cycles (T_{WAIT})** unless interrupted by door control buttons.
- **100 (CLOSING)**: The doors are closing. This process takes **at least 5 cycles**.

3. Direction Indicator

To allow passengers to know which direction the elevator is heading, each elevator must output the intended serving direction (`serve_dir`) for each elevator.

- When the elevator state is **OPEN**, `serve_dir` must accurately reflect the direction of the next movement (**01 for UP, 10 for DOWN**).
- Passengers will use this signal to determine whether to board. For example, if `serve_dir` is 01 (UP), only passengers whose destination is higher than the current floor will enter the elevator.



4. Door Control Buttons (Interrupts)

Passengers can manually interfere with the doors using `btn_open` and `btn_close`. The elevators must manage these signals dynamically:

If `btn_close` is pressed: If the door is OPEN, abort the remaining T_{WAIT} countdown and close the door, i.e., transition to CLOSING.

If `btn_open` is pressed: If the door is OPEN, reset the T_{WAIT} timer. If the door is CLOSING, abort the closing process and transition back to OPENING.

5. Overload Protection

A simulated weight scale tracks the exact number of people in each elevator, provided via `pax_cnt`.

The maximum capacity for each elevator is **15 passengers**.

When `pax_cnt` reaches 15, your design must assert the `is_full` output for that specific elevator.

When `is_full` is high, the elevator should operate in Bypass Mode. It can stop letting people out if there is a `car_req`, but should ignore external `up_req` / `down_req` to prevent boarding.

Inputs

Signal name	Bit width	Description
clk	1	Clock signal
rst_n	1	Asynchronous active-low reset signal
up_req	8	Indicates which floors have an active “Up” hall call, where bit 0 represents B1. The MSB is always low.
down_req	8	Indicates which floors have an active “Down” hall call, where bit 0 represents B1. The LSB is always low.
car_req	24	Concatenated 8-bit internal destination requests for each elevator, where bit 0 represents B1.
btn_open	3	Active-high signals representing the “Open Door” button press for each elevator
btn_close	3	Active-high signals representing the “Close Door” button press for each elevator
pax_cnt	12	Concatenated 4-bit current passenger counts for each elevator

Outputs

Signal name	Bit width	Description
cur_floor	9	Concatenated 3-bit current floor locations updated by the controller for each elevator 3'b000: B1 3'b001: 1F ... 3'b111: 7F
lift_status	9	Concatenated 3-bit operational states for each elevator 3'b000: IDLE 3'b001: MOVING 3'b010: OPENING 3'b011: OPEN 3'b100: CLOSING
serve_dir	6	Concatenated 2-bit indicators for the direction
is_full	3	Indicates whether each elevator has reached its maximum passenger capacity

Specifications

1. Top module name: **ECTRL** (File name: **ECTRL.sv**).
2. After asynchronous active-low reset, all output signals except for `cur_floor` (can be any value that is not unknown) must be reset to zero.
3. **No Teleportation:** For every elevator, `cur_floor` can **only** change when `lift_status` is MOVING. The floor number must be held for **at least 10 cycles**, and it can only **increase or decrease by 1** per move corresponding to the current `serve_dir`.
4. **Physical Constraints Check:**
For `lift_status`:
 - i. **IDLE:** The initial state. An elevator can be on **any** floor. It can transition to MOVING from IDLE.
 - ii. **MOVING:** It cannot directly transition to OPEN or CLOSING.
 - iii. **OPENING:** It can only either remain in its status or transition to OPEN after **5 or more cycles** have passed.
 - iv. **OPEN:** The elevator should maintain a timer to count for T_{WAIT} . If `btn_open` is detected, the timer should be restarted. It can transition to CLOSING only when the timer is up or `btn_close` is detected.
 - v. **CLOSING:** It cannot directly transition back to OPEN, and it should remain in its status for **5 or more cycles** before transition. However, if `btn_open` is detected, it should transition to OPENING immediately.For `serve_dir`:
 - i. **00** means idle. The elevator is not functioning. This **only** occurs in IDLE.
 - ii. **01** means going up; **10** means going down. **11** is an **invalid** value which must never occur.
 - iii. It can only change value when either in MOVING or in OPEN. (Or transition to these states.)
5. **Timeout Check:**
 - i. **Waiting Timeout:** Any passenger waiting at a hall must get in an elevator within **2000** cycles after their occurrence (spawn time). Otherwise, a timeout violation will be raised.
 - ii. **Boarding Timeout:** Any passenger who is already inside an elevator must reach their destination and get off within **2000** cycles after getting in. Otherwise, a timeout violation will be raised.(Note: This constraint ensures that your scheduling algorithm does not cause passenger starvation or trap passengers indefinitely.)

6. The 02_SYN result **must not have any errors** and **must not contain any latches**.
7. The slack in the timing report must be non-negative, and the result must be “Met.”
8. Gate level simulation **must not have any timing violation**.
9. Clock period is 10.0 ns.
10. Input delay = $0.5 \times$ clock period; output delay = $0.5 \times$ clock period.
11. Do not use `*error*`, `*latch*`, `*congratulation*` or `*fail*` as names for logic / wire / reg / submodule / parameter, otherwise the demo result will fail.
(Note: * represents any symbol before or after the word. E.g., error_test is prohibited.)
12. The demonstration will use **a hidden test pattern**. Ensure your design can handle various random traffic conditions, not just the test cases provided.
13. It is recommended to separate combinational blocks and sequential blocks.
14. **For loops are forbidden in this homework.**

Pattern Behavior

The occurrence of passengers, their waiting floors, and destinations are generated randomly. You may observe burst traffic (several hall requests within a few cycles) or extended idle periods (no requests for over a hundred cycles). A good scheduling algorithm will help improve the overall latency.

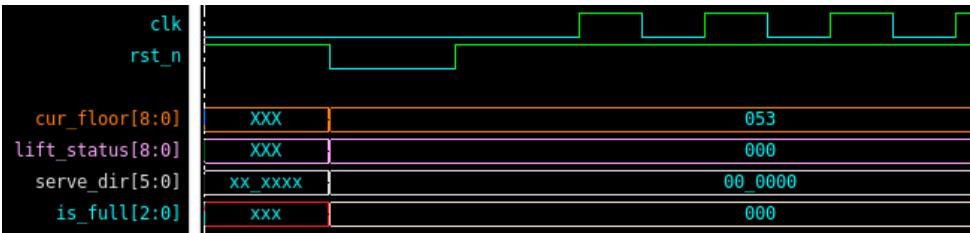
up_req / down_req keeps high until an elevator arrives in a certain direction. The hall requests fall **at the edge from OPEN to CLOSING**, and passengers will get in.

car_req keeps high until an elevator arrives. The car requests fall **at the edge from OPENING to OPEN**, and passengers will get off.

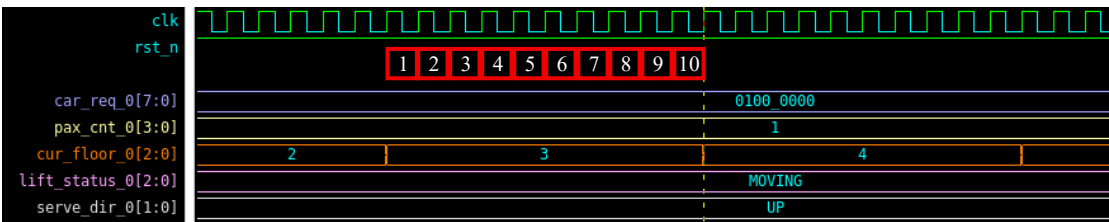
“Close Door” button will only be possibly pressed when OPEN; “Open Door” button will only be possibly pressed when OPEN or CLOSING.

Example waveform

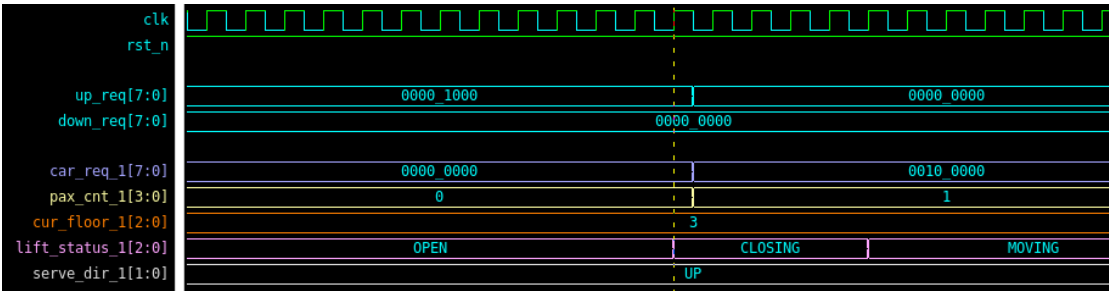
Reset all output after asynchronous and active-low reset signal.



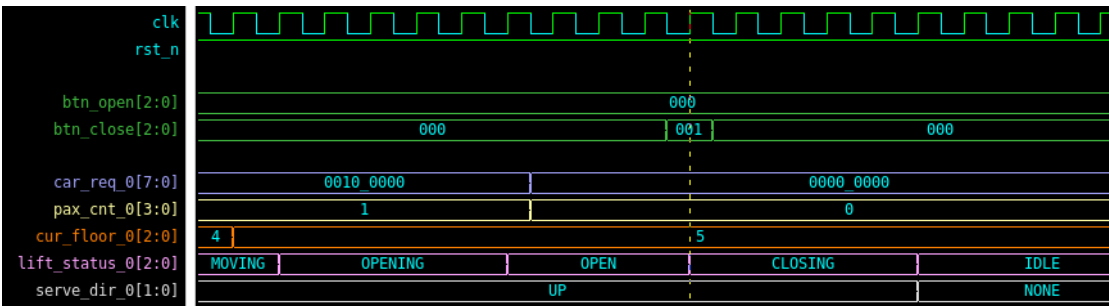
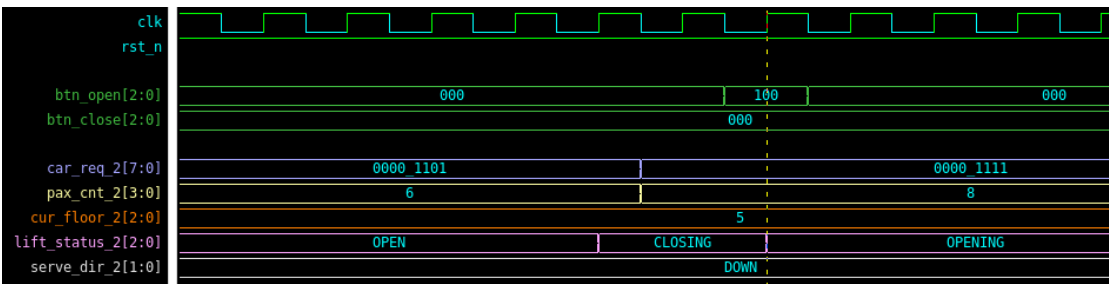
The movement of each elevator takes 10 cycles per floor.



An elevator responding to a hall request.



A button interrupts door operation.



Upload file

- Use `./01_upload` under `09_UPLOAD/` to submit your design & Submit your report to E3 (named `hw3_report_dcsXXX.pdf`).
- Note that the script requires time to synchronize with the NTP server. Please avoid last-minute submissions.
- 1de deadline: 4/16 23:59; 2de deadline: 4/21 15:30.

Grading policy

- Pass the RTL, Synthesis, and GATE level simulation: 60%.
- Performance: 30%
Figure of Merit (FoM): $\text{total latency} \times \text{area}^2$ (less is better)
(Latency for a passenger is defined as the duration between its occurrence time and its arrival time.)
- Report: 10%
- For submission in 2nd demo, total score will get **40% off**.

Note

Template folders and reference commands:

- `01_RTL/` (RTL simulation) → `./01_run_vcs_rtl`
- `02_SYN/` (synthesis) → `./01_run_dc`
- `03_GATE/` (gate-level simulation) → `./01_run_vcs_gate`
- `09_UPLOAD/` (design upload) → `./01_upload`